

Vedant Sarvesh Ranade

Boston, MA, 02115 | 339-213-0435 | ranade.v@northeastern.edu | github.com/vrmaverick | linkedin.com/vedantranade

EDUCATION

Northeastern University, Khoury College Of Computer Science

Boston, MA

Pursuing Master of Science in Artificial Intelligence, Concentration in Machine Learning

Sep. 2025 – Present

University of Mumbai

Mumbai, India

Bachelor's in Artificial Intelligence and Data Science; GPA 3.74/4.00

Nov. 2021 – May 2025

- Led as Class Representative for 2 years; Been part of a student committee's Technical and Sponsorship team

TECHNICAL SKILLS

Languages: Python, JavaScript, C, SQL(Postgres), ReactJS

Machine Learning: Scikit-learn, TensorFlow, PyTorch, Keras, Hugging Face, Ensemble learning, Bagging, Boosting

Deep Learning: CNN, Sequence Modeling, Timeseries Forecasting, Generative Models, OpenCV, Transformers, Regression

Natural Language Processing : LLM, VLM, TF-IDF, NER, LangChain, Ollama, AI Agents, Agentic RAG, Workflows

Data Science: Pandas, NumPy, Matplotlib, Seaborn, Data Cleaning, Feature Engineering, Data Mining, EDA, ETL

Deployment: Google Cloud Platform, Azure, Streamlit, LangSmith, Tensorboard, Git, FastAPI, Statistics, MCP

EXPERIENCE

Python Developer Intern

Jun. 2025 – Jul. 2025

Web creations, Dombivali, India

- Upgraded internal management portal, enabling real-time data visualization and easier Management using AI agents.
- Implemented 3D rendering models using transformers, solving a complex requirement saving costs by half.
- Optimized and proposed innovative workflows for large directories with architectural upgrades to improve scalability.

Web Developer Intern

Dec. 2023 – Dec. 2023

Web creations Dombivali, India

- Developed and maintained a JS-based backend automation and ensured efficient data management using SQL.
- Customized websites as per evolving project requirements, improving user experience, and demonstrating adaptability.
- Collaborated cross-functionally in design and development teams to accelerate project delivery and quality.

PROJECTS

Mint-Sage: AI Finance Manager | NLP, Pytorch, Streamlit, Pandas, Numpy

Oct. 2025 – Dec. 2025

- Built an AI finance assistant that automates categorization and forecasting to improve financial decisions.
- Engineered a MiniLM + SVM/XGBoost ensemble to classify transactions, enabling accurate financial analytics.
- Developed an ensemble model using sliding-window forecasting of the next 30-day expenses, outperforming other models.
- Applied kurtosis, skewness, and distribution drive normalization, helped in estimating distribution and improving stability.
- Designed an LLM budgeting layer that converts income, volatility, and forecast metrics into targeted recommendations.

EduBoost: AI Based Educational Tools | Transformers, ReactJS, Azure, FastAPI

Sep. 2024 – May 2025

- Researched and compared AI Architectures to enhance the performance of various supervised models.
- Developed a sketch-based math solver using custom CNN OCR and image processing techniques, for interpreting sketches.
- Fine-tuned Caffe model for illustration colorization, to achieve smoother and more realistic gradients to improve learning.
- Compared Transformer models BLIP and CLIP with modern LLMs to generate illustration summaries with safeguards.
- Designed a UI with ReactJS, Firebase and MongoDB, enabling seamless integration of models via FastAPI.
- Deployed Fine-tuned Caffe model using Azure, reducing the latency and increasing computational efficiency.

Deepfake Image Detection | Python, Flask, Azure, Tensorflow, OpenCV

Jan 2024 – May. 2024

- Architected a custom CNN model, using Tensorflow, achieving 92% accuracy in detecting AI-generated images.
- Built and deployed a Flask-based web application for real-time image verification, enabling more convenient interaction.
- Integrated and Deployed models using FastAPI and Azure, ensuring seamless integration and scalability.
- Pre-processed image dataset, using OpenCV and PIL, led to a robust model by reducing overfitting significantly.

AI Based Navigation System for Drones | Python, Unity, TCP Sockets, AI Algorithms

Sep. 2023 – Dec 2023

- Engineered an AI-driven drone navigation system in Unity, with a Python-based AI agent.

- Evaluated and benchmarked multiple AI algorithms across key performance metrics, enabling a comparative study.
- Established communication between the Unity simulation and Python backend using TCP sockets, reducing latency.
- Developed a Flask-based dashboard with automated SMTP notifications for a real-time Parcel Delivery app.

Medical assistance Chat-bot | *LangChain, Ollama, Streamlit*

Dec. 2025 – Jan. 2026

- Designed a Streamlit chatbot that lets users ask health questions and receive structured, medically grounded information.
- Built a LangChain RAG pipeline over medical descriptions and Q&A pairs so the chatbot can retrieve relevant context.
- Integrated Ollama-hosted OpenSource LLMs and embeddings to chunk and retrieve the data efficiently.
- Safety-focused prompting and disclaimers to guide possible conditions and next steps.

FinComplaints: Financial Complaints Management | *NLP, Streamlit, Tensorflow*

Oct. 2024 – Dec. 2024

- Built a Streamlit app to let companies search, filter, and analyze complaint tickets for faster issue triage.
- Cleaned and tokenized complaint corpus with custom NLP preprocessing techniques to remove noise for better modelling.
- Trained a TensorFlow text-classification model to route complaints into key product categories, helped sort as per departments.
- Added model predictions and confidence scores into UI, enabling traceability and transparency, in reviewing complaints.

Demand Based Dynamic Pricing Using AI | *Python, Scikit-Learn, Streamlit*

Jan. 2024 – May 2024

- Developed ensemble models with high dimensions, using SkLearn for demand forecasting and price optimization.
- Integrated real-time weather and temporal APIs based on location to dynamically adjust pricing strategies.
- Implemented invoice generation with UPI payment system through a Streamlit dashboard, streamlining the workflow.

Youtube Comment Analysis | *TensorFlow, Keras, TextBlob, Streamlit, Matplotlib*

May. 2025 – Jun. 2025

- Engineered an end-to-end NLP pipeline achieving 93% accuracy in intent classification of comments scraped using API.
- Integrated BiRNN-based intent classification and TextBlob sentiment analysis on YouTube comments.
- Automated comment Extraction and analyzed real-world comments and visualized sentiments using Matplotlib.
- Deployed a streamlit web app, enabling comment analysis in an interactive way.